CU Boulder Computer Science Capstone Project 25-26

Project Title

Ozone Interactive Story

Project Description

Develop an interactive, online, single-person activity based on the How Ozone Forms Near Earth's Surface in-person activity. This interactive activity will be made available online at the UCAR Center for Science Education website for use by the public and students and teachers in educational settings.

The How Ozone Forms Near the Earth's Surface in-person activity is a multi-student interactive activity designed as an interactive story. The intended audience is for upper elementary to middle school students. The online experience should be based off of the original activity but modified to be a single-player experience.

Learning objectives:

- Learners will understand how ozone pollution forms
- Learners will understand health impacts of ozone pollution
- Learners will discover actions that they can take to help reduce ozone pollution in their communities

The resulting single-player experience will be different from the in-person, multi-student experience. The team will discuss with the sponsors how to adapt the activity and the sponsors welcome creative ideas from the team. Some ideas include the following:

Part 1: What is ozone and how does it form?

A scene that contains all the elements of the story (based on the script from the How Ozone Forms Near Earth's Surface activity) with interactive elements for the user whenever possible. There could be audio "reading" the story as you go. Some content could be added as pop ups that the user clicks to reveal as they progress through the story. Or questions that are posed at different parts of the story to cue an interaction, such as:

- Choose the ingredients that must be in the air for ozone to form

- Make an ozone molecule.
- What are some ways that ozone pollution is harmful to our health?

Examples of the types of interactivity that could be included:

- Placing the ingredients needed to make ozone to a certain part of the screen
- Placing the Sun into the scene
- Placing an oxygen atom from the NO_2 and dropping it on the O_2 to create O_3 (ozone)
- Showing how ozone is higher in the summer (hot weather) than in the winter (cold weather)
- Click on the Sun to learn that sunlight activates chemical reactions in the air, when there's more sunlight, there's more reactions
- Click on the VOCs to learn a bit about them
- Click on the car/factory exhaust to learn about the pollutants that come from them
- Click on the kids playing outside to learn that ozone pollution can make it hard for them to breathe, and that you should stay indoors when ozone levels are too high
- Click on the houses that are near the factory to learn that people who live close to factories are exposed to more ozone pollution.

Part 2: Ozone Pollution Solutions

A new scene based on the Air Quality infographic (at the end of the activity), where the user can explore different solutions to ozone pollution. Click on different images to reveal text that explains the solution.

- Could categorize the solutions: which ones reduce the amount of sunlight (fueling/mowing in the evening), which ones reduce the amount of emissions (renewable energy, fuel efficient cars, walk, ride bikes, etc), which ones reduce the amount of VOCs (low VOC products, fuel efficient cars). This could be sorting the images by dragging them into different places, clicking on them to select them, or adding icons to each solution (a Sun icon, a VOC icon, a NO_2 icon)
- Could have the user select three (or one) solutions that they are excited to try out and then a "way to go" message such as: "Taking even just one of these actions helps to reduce the amount of ozone pollution and keep people safe" appears.
- Having an ozone meter that increases or decreases depending on different activities that you click on that affect ozone levels (eg using VOCs, car exhaust, factory pollution that increase ozone; pumping gas early morning or evening or riding your bike to decrease)

Project Requirements

The minimum requirements for this project are to develop a web application that provides a single-player interactive experience based on the How Ozone Forms Near the Earth's Surface activity. The interactive should include onboarding, tutorial features, and in-game prompts to scaffold learning and understanding.

The web application should be built using freely-available, open-source libraries and frameworks that can be supported long-term by UCAR staff. Angular is the strongly preferred web framework but UCAR and the development team will discuss and agree on the specific libraries and frameworks before development begins in order to ensure long-term maintainability of the final web application by UCAR staff. The application should be as efficient as possible to reduce server bandwidth and load times on the user's browser.

The web application should be accessible, browser responsive, and mobile-friendly so as to be appealing and usable to a wide variety of students who may be using a wide variety of devices. Application should be built from the start with principles of accessibility and usability in mind. This means that throughout the development lifecycle, the application should often be tested for:

- Keyboard accessibility
- Screen reader accessibility
- WCAG 2.2 AA compliance as much as possible
- Color blindness
- Mobile device accessibility and usability
- Usability

Students are expected to make intentional design choices that address accessibility and usability needs and discuss those choices, trade-offs, and decisions with the sponsors.

In the future, the interactive may be translated into other languages. With this in mind, students should design their code and data in a manner where the user can change languages. If Angular is used to develop the application, this can be accomplished by using the NGX-Translate module.

Deliverables

The primary deliverable is an accessible and mobile-friendly web application for the How Ozone Forms Near the Earth's Surface activity that uses free, open-source JavaScript libraries and web frameworks. The web application should be validated for proper coding techniques and accessibility periodically throughout the development lifecycle to ensure an accessible and bug-free final product. The web application should

be tested on a variety of devices to ensure usability across mobile devices and desktop browsers.

Code should be self-documenting and/or the development team should provide concise/accurate documentation to ensure long-term maintainability of the code by UCAR staff developers.

Ideally, students should implement unit testing and end-to-end testing. If Angular is used, students can use Karma, Jasmine, and Protractor to accomplish this.

The application should use the graphics provided by UCAR sponsors as well as the UCAR approved style guide for colors and fonts. Students can also incorporate their own custom graphics and animations, as needed.

Possible Extensions

While the in-person activity consists of static images, the students are welcome to explore avenues to include animation or other interactive components.

Skill Requirements

Web application design and development

- Familiarity with web application development using HTML, CSS, and Javascript.
- Familiarity with common Javascript frameworks such as Angular.
- Familiarity with using web APIs and libraries to potentially generate graphics, animations, or audio such as particles.js, tone.js, PIXI.js. Students can explore options but should discuss ideas with sponsors.
- Familiarity with user interface design and user experience design.
- Familiarity with mobile-friendly and responsive design.
- Familiarity with developing accessible websites.
- Familiarity with Github and code repositories

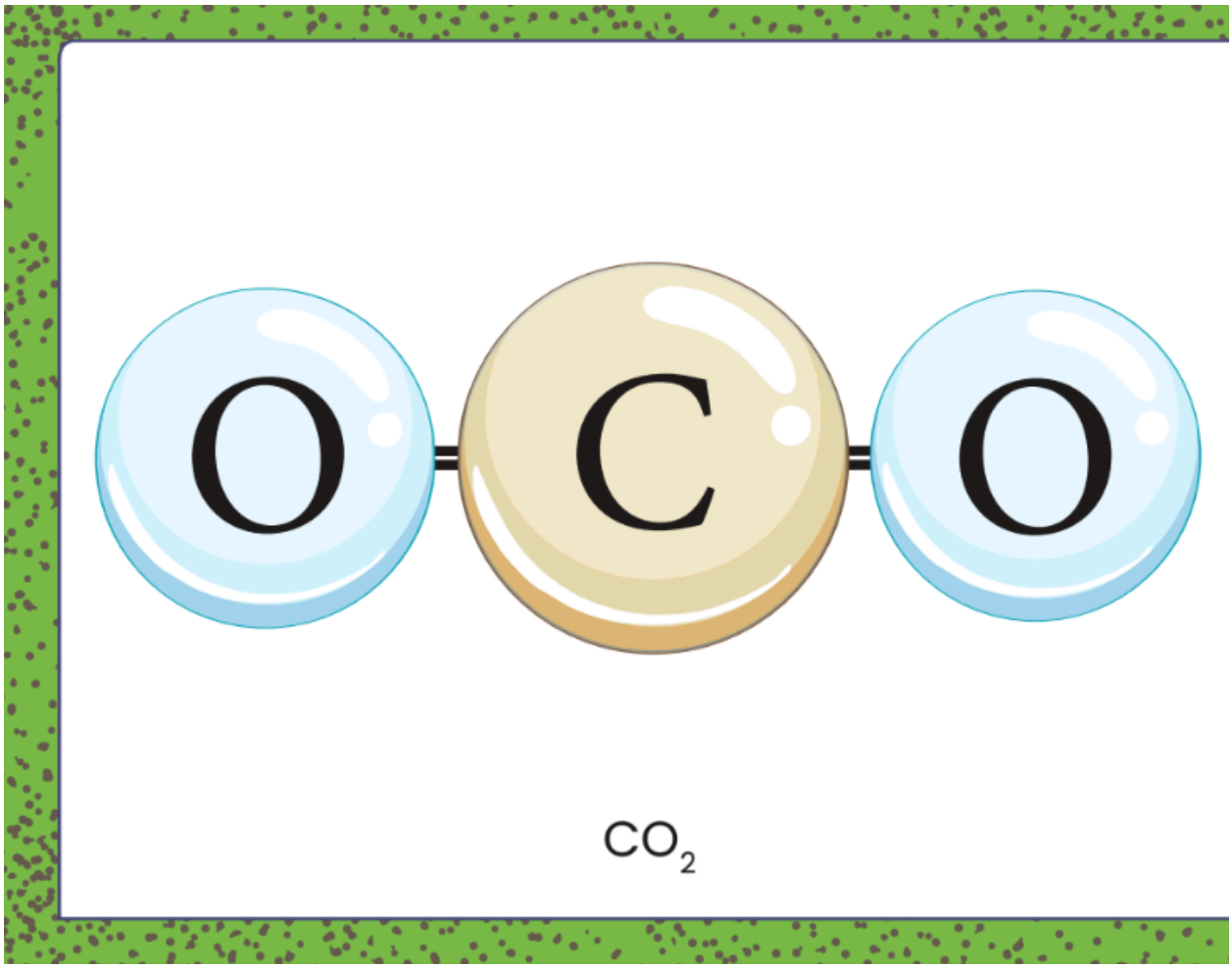
Optional Skills:

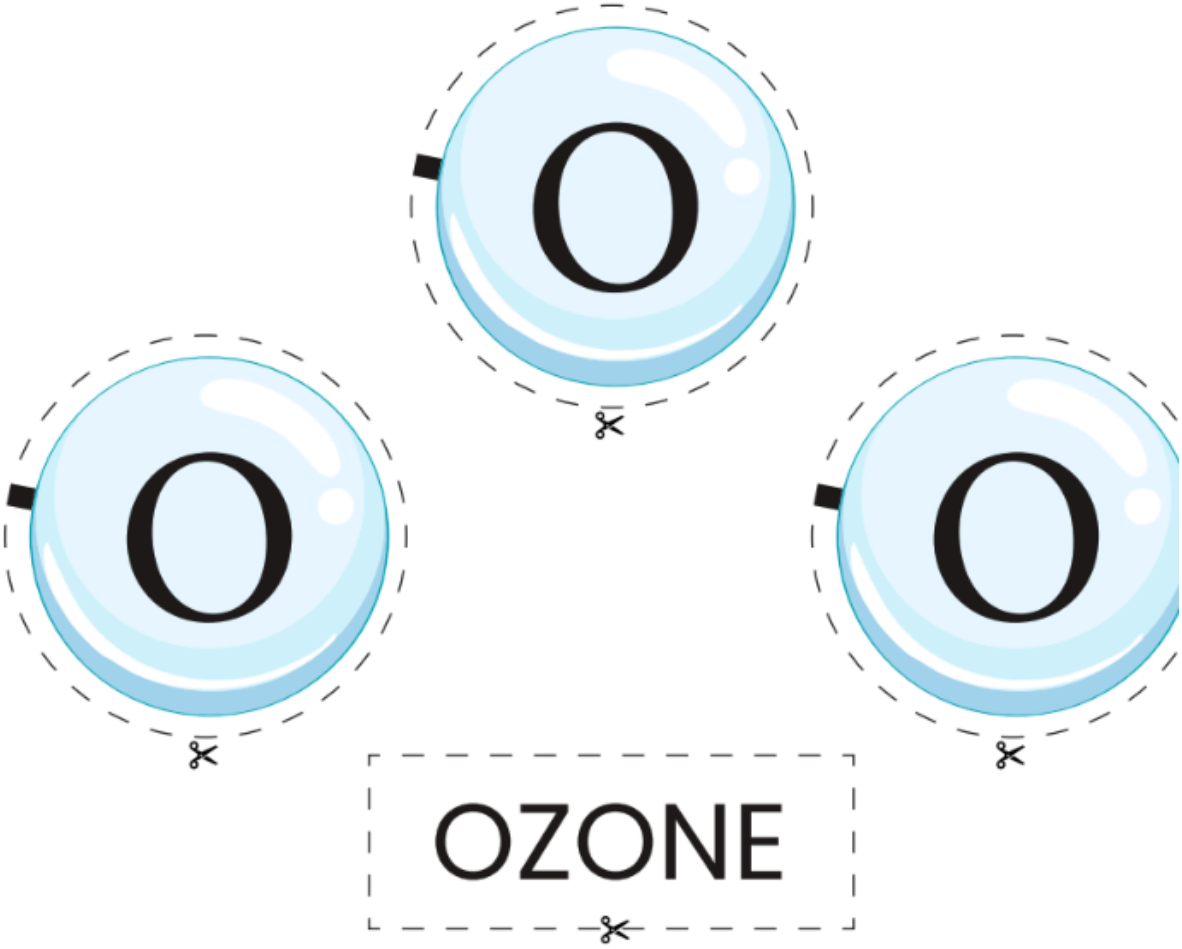
We welcome students with an interest in usability who can conduct usability tests throughout the development lifecycle. This is a great opportunity to iterate on design options and prototype with potential users before extensive development effort is committed.

Available Resources and Documents

- How Ozone Forms Near the Earth's Surface activity

Example Images:

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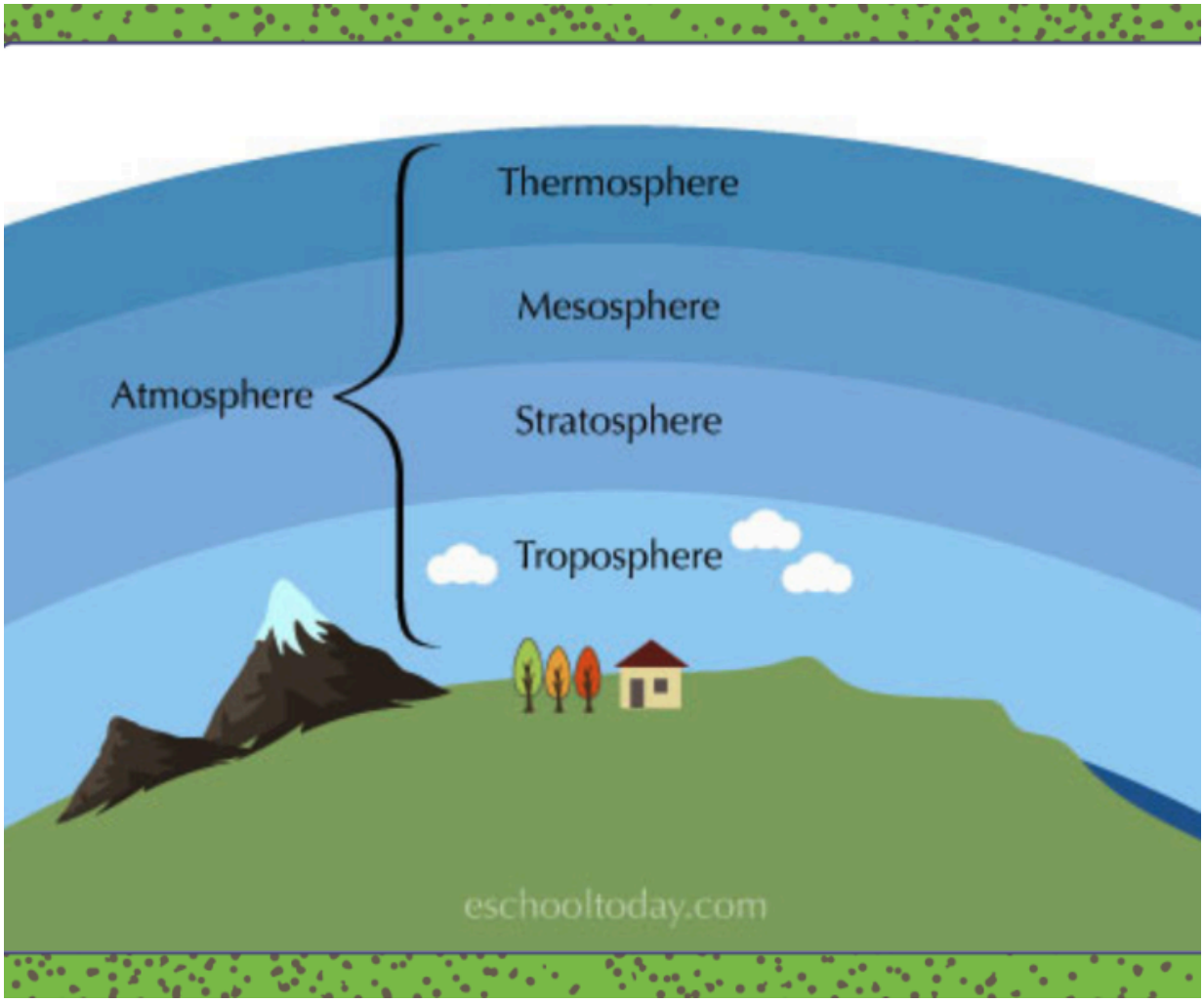
VOC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Mesosphere
Atmosphere
Stratosphere
Troposphere
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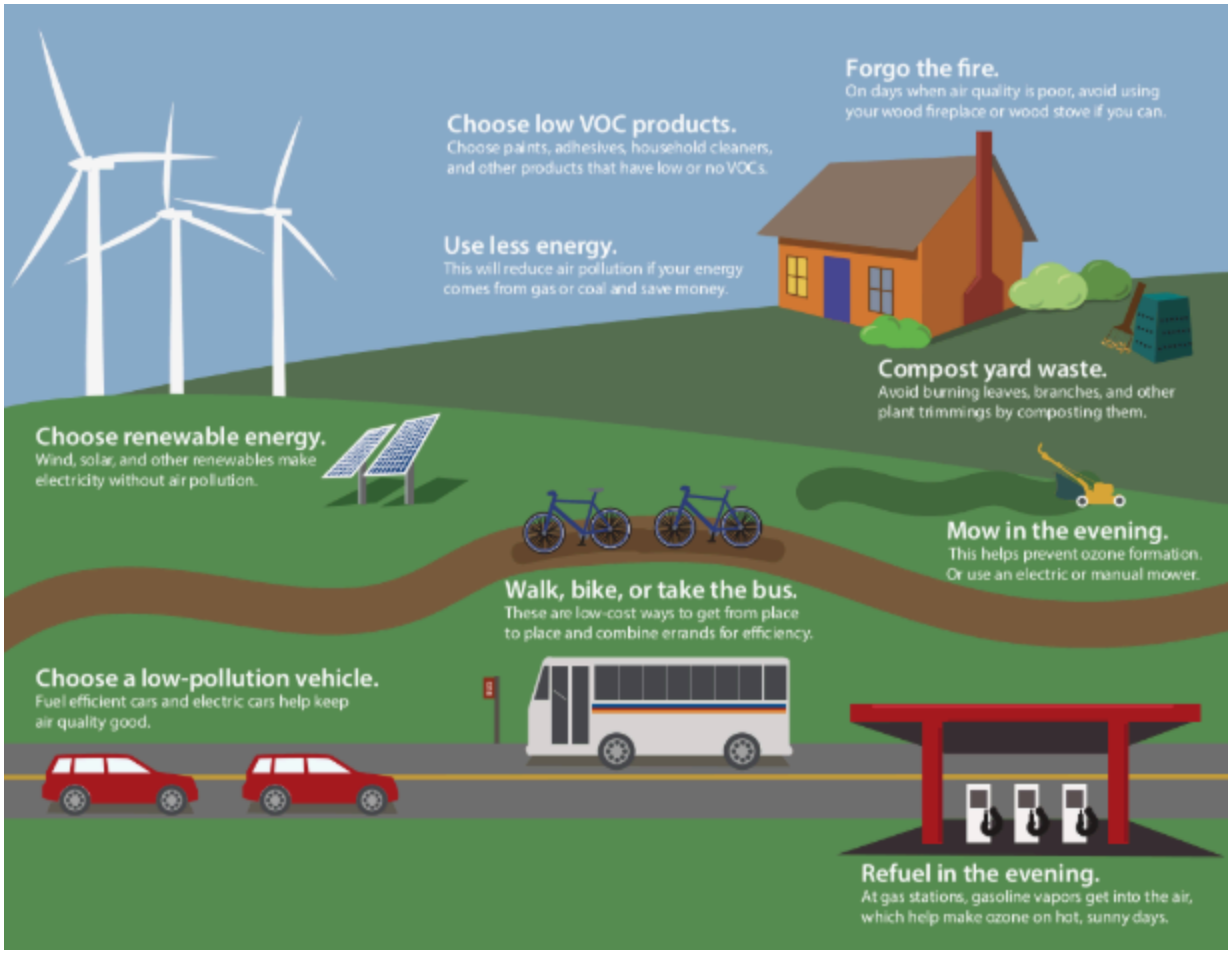
Forgo the fire.
On days when air quality is poor, avoid using your wood fireplace or wood stove if you can.
Choose low VOC products.
Choose paints, adhesives, household cleaners, and other products that have low or no VOCs.
Use less energy.
This will reduce air pollution if your energy comes from gas or coal and save money.
Compost yard waste.
Avoid burning leaves, branches, and other plant trimmings by composting them.
Choose renewable energy.
Wind, solar, and other renewables make electricity without air pollution.
Mow in the evening.
This helps prevent ozone formation. Or use an electric or manual mower.
Walk, bike, or take the bus.
These are low-cost ways to get from place to place and combine errands for efficiency.
Choose a low-pollution vehicle.
Fuel efficient cars and electric cars help keep air quality good.
Refuel in the evening.
At gas stations, gasoline vapors get into the air, which help make ozone on hot, sunny days.